

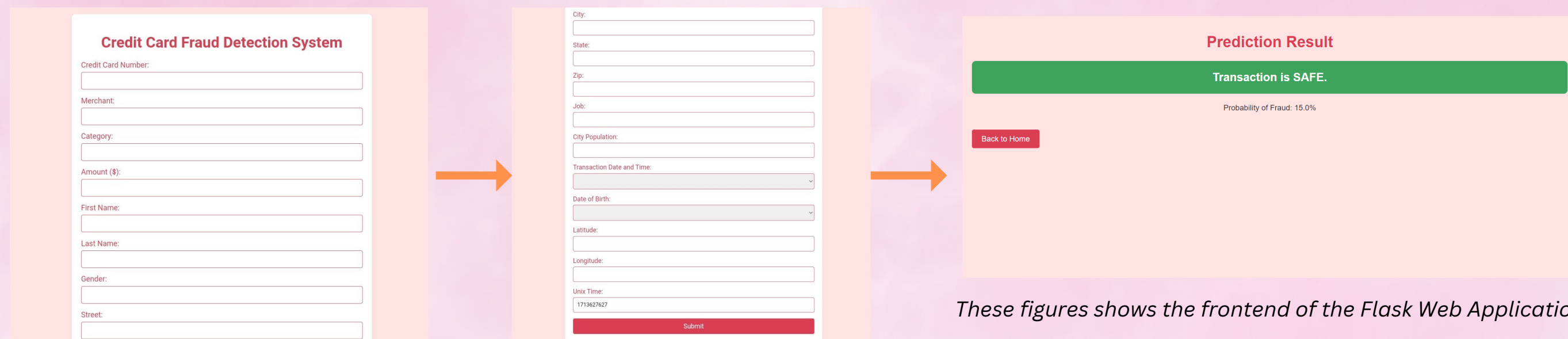


Using Machine Learning

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These figures shows the frontend of the Flask Web Application

Introduction:

The surge in electronic transactions has amplified vulnerabilities to credit card fraud. This project leverages machine learning to create a real-time detection system, employing a RandomForest classifier within a Flask application to enhance transaction security.

Project Objectives:

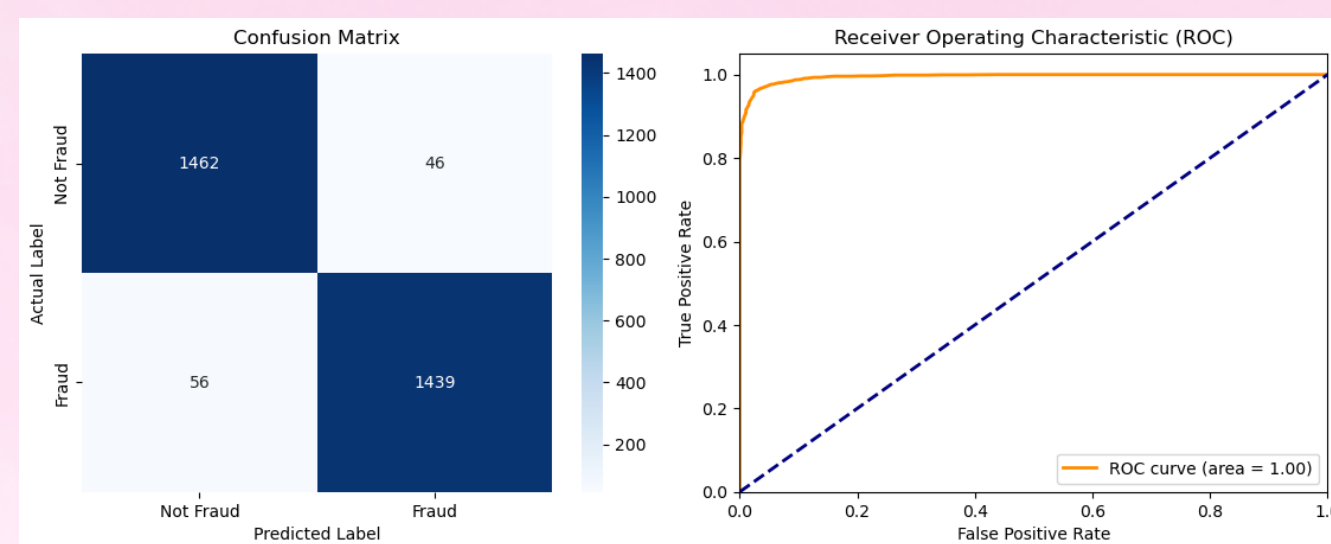
- Conduct in-depth analysis of credit card transaction data.
- Develop and train a RandomForest classifier to predict fraudulent transactions.
- Integrate the classifier into a real-time application.
- Validate model performance against industry-standard metrics.

Methodology:

- Data preprocessing included normalization and encoding.
- Model selection process involved evaluating multiple algorithms.
- Chose RandomForest due to its effectiveness with imbalanced data.
- The model was trained, validated, and tested for accuracy, precision, recall, F1 score, and AUC.

Results:

Model	Accuracy	Precision	Recall	F1 Score	AUC
Random Forest	0.95	0.92	0.89	0.90	0.94
Gradient Boosting	0.94	0.90	0.88	0.89	0.93
Logistic Regression	0.89	0.85	0.83	0.84	0.88



This figure shows an ROC curve with a perfect AUC of 1.00, and the accompanying confusion matrix demonstrates the model's accuracy in distinguishing between fraudulent and non-fraudulent transactions.

Evaluation:

Evaluated based on precision, recall, and F1-score. The system was user-tested for real-world applicability, confirming the model's high accuracy and user-friendly interface.

Conclusions:

The project successfully meets its objectives, proving that machine learning can significantly enhance fraud detection capabilities in financial transactions, thus reducing fraudulent losses and improving security.

Limitations and Further Work:

Future enhancements include real-time performance optimization, adaptation to new fraud patterns, and exploration of deep learning models.

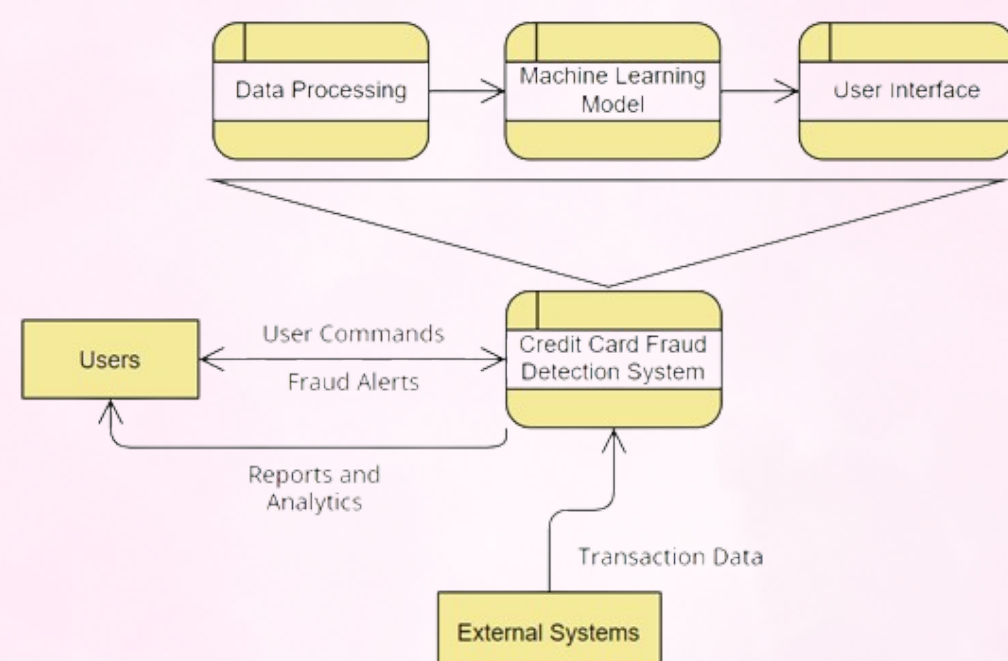
References:

- 1.R. J. Bolton and D. J. Hand, "Statistical fraud detection: A review," Statistical Science, vol. 17, no. 3, pp. 235-249, 2002.
- 2.A. Dal Pozzolo, G. Boracchi, O. Caelen, C. Alippi, and G. Bontempi, "Credit card fraud detection: A realistic modeling and a novel learning strategy," IEEE Transactions on Neural Networks and Learning Systems, vol. 29, no. 8, pp. 3784-3797, Aug. 2015.

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This figure illustrates the system architecture for the credit card fraud detection system, detailing the flow from data processing and machine learning model to user interface, and the interaction with users and external systems.